

Skills Summary

Precise Definitions and Counterexamples

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading a definition exactly

A definition must be testable. Every word has to name something you can check on a figure: a location (“on the segment”), a relation (“congruent”), or a measure (“ 90° ”).

Words that are not conditions: “looks like”, “about the same”, “roughly”. If a definition rests on one of these, it cannot be used in a proof.

Perpendicular lines: two lines that intersect *to form a right angle*. The angle clause is the whole content — “two lines that intersect” is every crossing pair.

Example: Lines j and k are perpendicular, and one of the four angles they form measures $(4x + 10)^\circ$. Find x .

Step 1. The definition of perpendicular says the angle formed is a right angle, so I can replace the word “perpendicular” with the equation “that measure is 90° ”.

Step 2. $4x + 10 = 90$, so $4x = 80$.

$$\Rightarrow x = 20$$

Try it: Lines p and q are perpendicular, and one angle they form measures $(6x - 12)^\circ$. Find x .

check: $12 = x$

Skill 2 — Turning a definition into an equation

Midpoint of $\overline{AB}$: the point M on $\overline{AB}$ with $\overline{AM} \cong \overline{MB} \Rightarrow AM = MB$.

Angle bisector of $\angle ABC$: the ray $\overrightarrow{BD}$ with $\angle ABD \cong \angle DBC \Rightarrow m\angle ABD = m\angle DBC$.

Congruent segments: $\overline{PQ} \cong \overline{RS} \Rightarrow PQ = RS$.

Each definition turns one congruence mark into one equation. Write the equation first, then solve it.

Example: M is the midpoint of $\overline{AB}$, with $AM = 4x + 3$ and $MB = 6x - 9$. Find x .

Step 1. Midpoint means the two pieces are congruent, and congruent segments have equal lengths — so the definition hands me the equation $AM = MB$.

Step 2. $4x + 3 = 6x - 9$, so $12 = 2x$.

$$\Rightarrow x = 6$$

Try it: $\overrightarrow{BD}$ bisects $\angle ABC$, with $m\angle ABD = 5x - 2$ and $m\angle DBC = 3x + 8$. Find x .

check: $5 = x$

Skill 3 — Building a counterexample

To disprove “if P then Q ”, produce one case that **satisfies** P and **fails** Q . One case is enough, and it must do both jobs — always check the hypothesis numerically before claiming the conclusion fails.

Example: Disprove: “If two angles are supplementary, then each one is acute.” Two angles measure a and b degrees.

Step 1. I need a pair that meets the hypothesis — supplementary means the measures add to 180 — and breaks the conclusion, so at least one must be 90 or more; try $a = 120$ and $b = 60$.

Step 2. Hypothesis check: $a + b = 120 + 60$, and 120 is obtuse, so the conclusion fails.
 $\Rightarrow$ the sum is **180**, so the pair really is supplementary, and the claim is false.

Try it: Disprove “if two angles are complementary, then they are congruent” with the pair $a = 30$ and $b = 60$: check the sum the hypothesis needs.

06 :check

(!) **Watch out:** a counterexample that fails the “if” part proves nothing. Check the hypothesis first, then show the conclusion breaks.